NCBI BIOLOGICAL DATASET SELECTION

Name: Shafaque Naushad

Course: B.Sc. Life Science (Pursuing)

Project Type: Individual Bioinformatics Project

DATASET INFORMATION:s

Dataset Name: A single cell and spatial atlas of preinvasive Breast cancer [Visium ST]

NCBI GEO accession number: GSE348919

Repository: NCBI GEO (Gene Expression Omnibus)

Organism: Homo sapiens (Human)

Data Type: Spatial transcriptomics (Gene expression)

Number of Samples: 14

Platform: GPL24676- Illumina NovaSeq 6000

Bio Project Accession: PRJNA1536776

Status: Public on Sep 29, 2026

Raw Data: Available on SRA

Supplementary File: GSE348919_RAW.tar

OBJECTIVE:

To select and study the single cell and spatial atlas of preinvasive breast cancer [Visium ST] with accession number- GSE348919 from NCBI and formulate research question, and understand dataset and its research relevance.

RESEARCH QUESTIONS:

Question 1: Why does some DCIS remain localised while other DCIS progress toward invasive breast cancer?

Answer: Some DCIS remains localised because the abnormal cells have not adapted all the changes to invade. Other cells may have acquired or developed genetic changes to interact and invade nearby cells to cause tumour.

Question 2: How do DCIS cells interact with surrounding cells within the tumour microenvironment and what are their effect on human body?

Answer: DCIS cells interact with surrounding cells, including immune cells, fibroblasts and other cells present in breast tissue**.** It can cause inflammation, suppress normal growth of cell, and lead to cause tumour. If DCIS becomes invasive, cancer cells can enter surrounding breast tissue and can increase rapidly. However, the effects vary by person to person.

Question 3: What is the major difference between DCIS and invasive breast cancer?

Answer: The major difference between DCIS and invasive breast cancer is: In DCIS, abnormal cell remains inside the breast milk ducts and have not invaded surrounding tissue. In invasive breast cancer cell have crossed the duct and spread near surrounding tissue and may spread to other body parts.

Question 4: Can molecular and spatial characteristics of DCIS help in identifying potential treatment targets?

Answer: Yes, molecular and spatial characteristics of DCIS help in identifying potential treatment targets. Gene expression data can show which genes are active in different tissue regions, while spatial transcriptomics helps reveal the locations of those regions and their relationships with nearby cells.

DATASET DESCRIPTION:

NCBI Gene Expression Omnibus (GEO) with accession number GSE348919. Titled as “A single cell and spatial atlas of preinvasive breast cancer [Visium ST].” The disease being studied is ductal carcinoma in situ (DCIS). It tells about cancerous cell present in milk duct which have not invaded the surrounding tissue. Spatial transcriptomics helps identifying which genes are active and expressing and where their expression occurs within the tissue. It helps to understand the difference between DCIS, normal breast tissue, invasive breast cancer and the tumour microenvironment. The database contains 14 samples analysed on Homo sapiens which makes easy to understand gene expression, spatial patterns, tissue/cell differences and biological pathways. Each sample has its own identifier and represents data obtained from a particular tissue.

DATASET SIZE:

Number of samples: 14

Raw data file size: 583.8MB

Raw data file: GSE348919_RAW.tar

CONCLUSION:

This dataset provides information about DCIS and its surrounding tissue. Spatial transcriptomics helps in understanding gene expression and its location within tissue. The dataset difference between DCIS, normal breast tissue, invasive breast cancer and the tumour microenvironment. The database contains 14 samples from Homo sapiens.

Reference / Source:

NCBI Gene Expression Omnibus (GEO),

GSE348919

https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE348919